

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2106

Kalsen Gypsum Plant 138 kV
February 29, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2106. The Interconnection Customer proposes a 355.7 MW solar photovoltaic facility interconnecting at the Kalsen Gypsum Plant 138 kV substation in the Delaware area. The request was received on June 08, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 540 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,918,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: G.E. Control / Kalsen Gypsum Plant 138 kV L
Contingency: White Bluff / Orange Bulk Substation 138 kV
Loading: 208.6% (Rating: 195 MVA)
DFAX: 77.7%
Pre-project loading: 67.2%
2. Facility: G.E. Control / Wheatland Station 138 kV
Contingency: Orange Bulk / White Bluff 138 kV
Loading: 190.5% (Rating: 209 MVA)
DFAX: 77.7%
Pre-project loading: 58.4%
3. Facility: White Bluff / Kalsen Gypsum Plant 138 kV
Contingency: G.E. Control / Kalsen Gypsum Plant 138 kV
Loading: 150.2% (Rating: 213 MVA)
DFAX: 100.0%
Pre-project loading: 16.7%
4. Facility: Warren F Sam Beasley Generation Station Substation / Van Sant ta 138 kV Line
Contingency: Wheatland Sta / Mckee Run 138 kV
Loading: 120.3% (Rating: 212 MVA)
DFAX: 29.1%
Pre-project loading: 71.4%
5. Facility: Gilroy Food / Main Substation 138 kV
Contingency: Mckee Run / Seneca 138 kV
Loading: 109.7% (Rating: 195 MVA)
DFAX: 11.7%
Pre-project loading: 88.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
G.E. Control / Kalsen Gypsum Plant 138 kV L	Rebuild line with higher-capacity conductor	\$7,174,000	75% (\$5,361,000)
Wheatland Sta / G.E. Control 138 kV Line	Rebuild line with higher-capacity conductor	\$10,388,000	75% (\$7,763,000)
White Bluff / Kalsen Gypsum Plant 138 kV	Rebuild line with higher-capacity conductor	\$5,109,000	100% (\$5,109,000)
Warren F Sam Beasley Generation Station Substation / Van Sant ta 138 kV Line	Rebuild line with higher-capacity conductor	\$8,874,000	87% (\$7,688,000)
Gilroy Food / Main Substation 138 kV	Reconductor line	\$2,474,000	40% (\$997,000)
	Total allocated to project		\$26,918,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.